Homework 2 – k Sample Inference

1 Question 1: Why do we Correct for Multiple Comparisons?

A student asked me what the point of correcting for multiple comparisons was. They made a two interesting points:

1. They are different outcomes, so, unlike the coin flip example (where the coin is the same) it doesn't makes sense to combine them.
2. If a manuscript presented three studies with t-tests, they don't correct for multiple comparisons across *those* tests.

When we conduct all pairwise comparisons after an omnibus test like ANOVA, we are testing a single family of hypotheses to answer one overarching question ("Which of these groups are different?"). Every additional test provides another opportunity to commit a Type I error from random noise. Similar to the coin, the probability of making at least one false error across J independent tests grows exponentially: $1-(1-\alpha)^J$. Note that our pairwise comparisons are not independent, so it's slightly different.

We don't correct for inferences across separate studies because they typically test distinct hypotheses using independent samples. Our attempt to address the multiple comparisons problem is specifically confined to the issue of increased Type I error within a single decision-making context. If we don't, we can find ourselves in a p-hacking-like situation, where an investigator might search through many tests to find one $p < 0.05$ (a potential false discovery), rather than making decisions to best position themselves for finding a true discovery.

Here, we will quickly demonstrate why we correct for the multiple comparisons problem when conducting several tests to answer the same question.

1.1 Part a

Complete the following steps in $R = 1000$ simulations.

1. Simulate data for $k = 5$ samples $n = 30$ drawn from a standard Gaussian distribution.
2. Use `pairwise.t.test(..., p.adjust.method="none")` to compute unadjusted pairwise t-tests and record if any comparison yields statistically discernible support for a non-zero mean.
3. Use `rstatix::tukey_hsd(...)` to compute the Tukey's HSD adjusted pairwise tests and record if any comparison yields statistically discernible support for a non-zero mean.

In the end, describe the Type I error rates.

```r
set.seed(1212)

k <- 5    # number of samples
n <- 30   # sample size
R <- 1000 # simulations

group <- factor(rep(1:k, each = n))

sig_none <- logical(R)
```

```r
sig_tukey <- logical(R)

for (i in 1:R) {

  value <- rnorm(k * n, mean = 0, sd = 1)
  sim_data <- tibble(group = group, value = value)

  sig_none[i] <- any(pairwise.t.test(x = sim_data$value,
                                     g = sim_data$group,
                                     p.adjust.method = "none")$p.value < 0.05,
                     na.rm = TRUE)

  sig_tukey[i] <- any(tukey_hsd(aov(value ~ group, data = sim_data))$p.adj < 0.05)

}

# Type 1 error rates
mean(sig_none)
```

```
[1] 0.279
```

```r
mean(sig_tukey)
```

```
[1] 0.049
```

The Type 1 error rate for the uncorrected test is 27.9%, while the Type 1 error rate for the corrected test is 4.9%, which is close to 5%. This makes sense because the corrected test caps the FWER at $\alpha = 0.05$. The uncorrected test has a much higher chance of detecting a difference when there is none (detecting noise) than the corrected test.

1.2 Part b

Redo the simulation, changing to adjust according to the false discovery rate `p.adjust.method="BH"` and comment on the differences.

```r
set.seed(1212)

k <- 5    # number of samples
n <- 30   # sample size
R <- 1000 # simulations

group <- factor(rep(1:k, each = n))

sig_bh <- logical(R)
sig_tukey <- logical(R)

for (i in 1:R) {

  value <- rnorm(k * n, mean = 0, sd = 1)
  sim_data <- tibble(group = group, value = value)

  sig_bh[i] <- any(pairwise.t.test(x = sim_data$value,
                                     g = sim_data$group,
                                     p.adjust.method = "BH")$p.value < 0.05,
```

```
                      na.rm = TRUE)

  sig_tukey[i] <- any(tukey_hsd(aov(value ~ group, data = sim_data))$p.adj < 0.05)

}

# Type 1 error rates
mean(sig_bh)
```

```
[1] 0.039
```

```
mean(sig_tukey)
```

```
[1] 0.049
```

The BH correction brought the Type 1 Error rate down a substantial amount to 0.039, towards $\alpha = 0.05$ like the Tukey test. I notice that the BH correction is slightly lower than the Type 1 error rate for the Tukey test which was 0.049, but they are much closer than the simulation in Question 1.1. The reason that they might not be the same is because BH limits the proportion of false positives to 5% while the Tukey test limits FWER and the probability of making 1 Type 1 error at all to 5%, so they correct for different things.

2 Question 2: Power Analyses

Revisit the MFAP4 study from class.

2.1 Part a

Assume the researchers were targeting a moderate effect size of Cohen's $f = 0.25$, setting significance level $\alpha = 0.05$ and desired power $1 - \beta = 0.80$. Were they adequately powered?

Note: The actual effect size is $f = 0.55$, but researchers usually do not know that ahead of time and defaulted to moderate, instead of specifying a large effect.

What is the required sample size per group (n) and the overall total sample size (N)? Write the `R` code using the `pwr` package to calculate this sample size. This package assumes all samples are of the same size so $N = nk$, where k is the number of samples. Note that the best answers will have a corresponding graphical display.

```
k <- 5         # Number of groups
f.mod <- 0.25  # moderate effect size using Cohen's f
alpha <- 0.05  # significance level

(n.small <- ceiling(pwr.anova.test(k = k,
                                   f = f.mod,
                                   sig.level = alpha,
                                   power = 0.80)$n))   # sample size per group
```

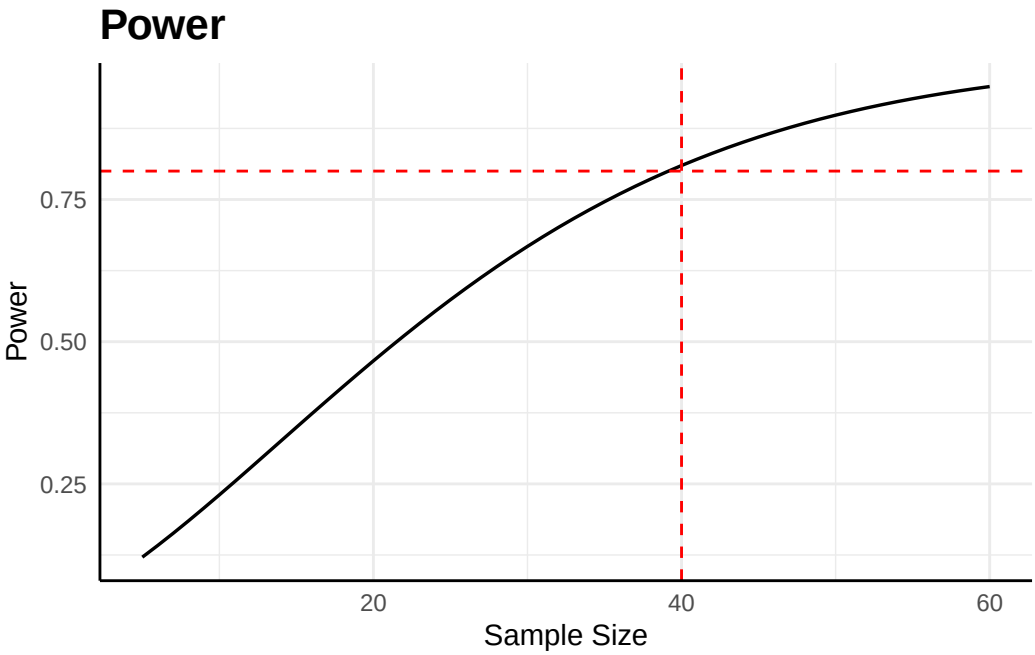
```
[1] 40
```

```
N.small <- n.small * k   # total sample size

# Power curve across a range of n
power_curve <- tibble(n = seq(5, 60, by = 1)) |>
```

```r
  mutate(power = map_dbl(n, ~ pwr.anova.test(k = k, n = .x, f = f.mod,
                                             sig.level = alpha)$power))

ggplot(power_curve, aes(x = n, y = power)) +
  geom_line(linewidth = 0.6) +
  geom_hline(yintercept = 0.80, linetype = "dashed", color = "red") +
  geom_vline(xintercept = n.small, linetype = "dashed", color = "red") +
  labs(
    x = "Sample Size",
    y = "Power",
    title = "Power"
  ) +
  theme_minimal() +
  theme(
    axis.line = element_line(color = "black"),
    plot.title = element_text(face = "bold", size = 16)
  )
```



The researchers need a sample size of at least n = 40 per group to achieve 80% power, so the MFAP4 study is adequately powered because the lowest group sample size was 67 (> 40). The total sample size needed is N = 200.

2.2 Part b

The project was funded by several agencies. Suppose that a funding agency suggested funding limiting the researchers to $N = 150$. Evaluate this suggestion in the context of the ANOVA. Would that be large enough to be adequately powered?

Compute the non-centrality parameter

$$\lambda = k \cdot n \cdot f^2$$

and use the non-central F distribution to compute the power under H_a for Cohen's $f = 0.25$. Calculate the statistical power achieved under this budget constraint. Note that the best answers will have a corresponding graphical display.

```r
k <- 5          # Number of groups
n <- 30         # sample size per group
f <- 0.25       # Cohen's f effect size
alpha <- 0.05  # significance level

(pwr.anova.test(k = k,
                n = n,
                f = 0.25,
                sig.level = alpha)$power)
```

```
[1] 0.6675964
```

```r
# non-centrality parameter for plotting
lambda <- k * n * f^2

df1 <- k - 1
df2 <- k * (n - 1)
F_crit <- qf(1 - alpha, df1, df2) # F-statistic

# x-axis range wide enough to show both curves fully
x_max <- qf(0.999, df1, df2, ncp = lambda) * 1.1
x <- seq(0, x_max, length.out = 1000)

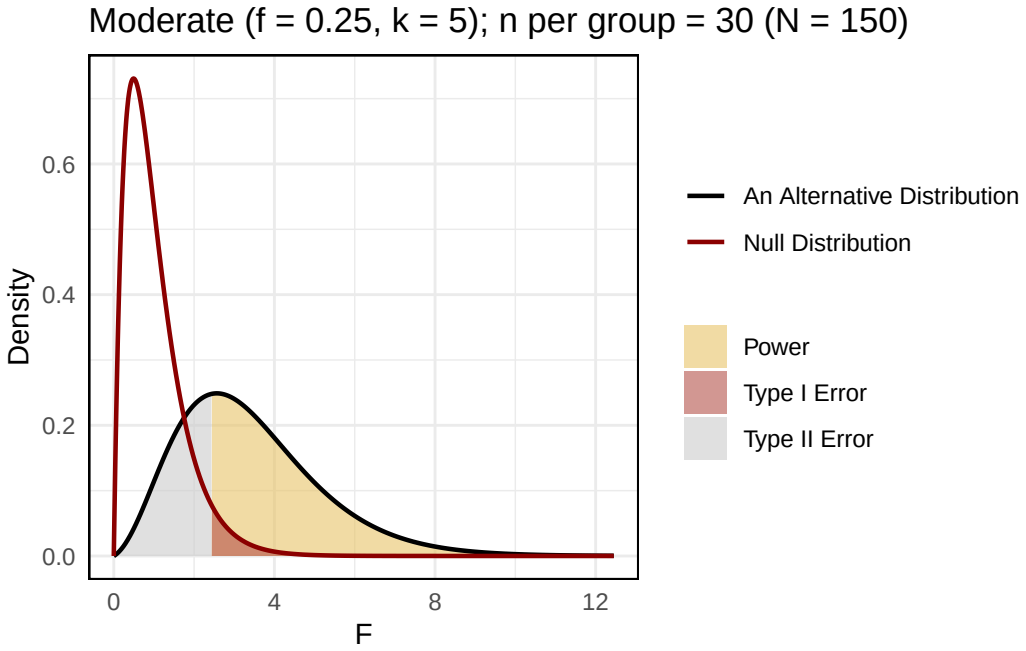
dat <- tibble(
  x = x,
  null = df(x, df1, df2),
  alt  = df(x, df1, df2, ncp = lambda)
)

# Shaded regions
type_ii <- dat |>
  filter(x <= F_crit) |>
  transmute(x, y = alt, region = "Type II Error")
power   <- dat |>
  filter(x >= F_crit) |>
  transmute(x, y = alt, region = "Power")
type_i  <- dat |>
  filter(x >= F_crit) |>
  transmute(x, y = null, region = "Type I Error")

regions <- bind_rows(type_ii, power, type_i) |>
  mutate(region = factor(region, levels = c("Power", "Type I Error", "Type II Error")))

ggplot() +
  geom_area(data = regions, aes(x = x, y = y, fill = region),
            alpha = 0.6, position = "identity") +
  geom_line(data = dat, aes(x = x, y = alt, color = "An Alternative Distribution"),
            linewidth = 0.8) +
  geom_line(data = dat, aes(x = x, y = null, color = "Null Distribution"),
            linewidth = 0.8) +
  scale_fill_manual(values = c("Power" = "#E8C468",
                               "Type I Error" = "#B5524A",
                               "Type II Error" = "grey80")) +
  scale_color_manual(values = c("An Alternative Distribution" = "black",
                                "Null Distribution" = "darkred")) +
  labs(
```

```r
    title = paste0("Moderate (f = ", f, ", k = ", k,
                   "); n per group = ", n, " (N = ", n * k, ")"),
    x = "F",
    y = "Density",
    fill = NULL,
    color = NULL
  ) +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.7),
    legend.position = "right"
  )
```



The statistical power with N = 150 from the budget constraint is around 67%, which does not meet the general rule of 80%.

2.3 Part c

In the article the researchers state:

> *However, the statistical power was not sufficient to allow an accurate discrimination of no to moderate stages of hepatic fibrosis (F0–F2) from severe fibrosis and cirrhosis (F3 and F4).*

The European Association for the Study of the Liver (EASL) provides treatment recommendations based on these stages.

- Stages 0-1: Treatment is deferred with regularly assessment
- Stage 2: Treatment is justified for those in stage 2
- Stage 3-4: Treatment is a priority for those in stages

Suppose the researcher wanted to perform a follow-up test where they want to ensure adequate power not just for the overall ANOVA F-test, but specifically to detect a pairwise differences between stages 1 and 2. Explain why the sample size computed in Part a may not be enough and use `PMCMRplus::power.tukey.test()` to compute a sample size that attains adequate power.

Note: The researchers would have had to do this *a priori*. Because we aren't medical experts, you can use sample estimates using the original data.

$$
\begin{aligned}
n &= \text{answer from part a} \\
\mu_\Delta &= 0.25 \\
\sigma^2_{\text{pooled}} &= \frac{(176-1)0.85^2 + (135-1)0.84^2}{176+135-2} = 0.7152 \\
\sigma_{\text{within}} &= 0.77
\end{aligned}
$$

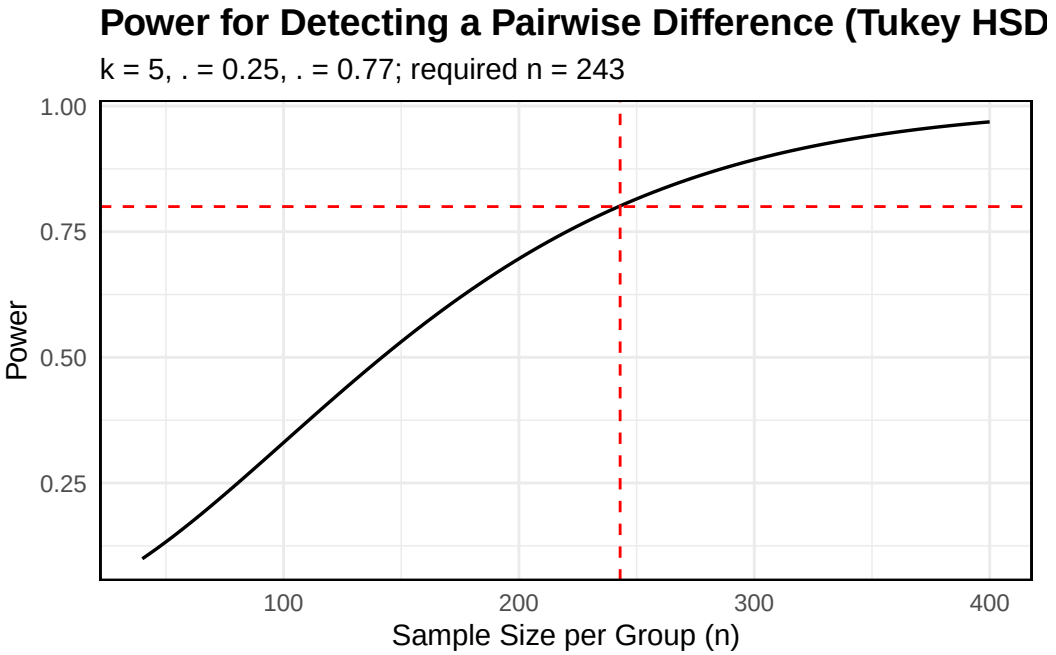
```r
# parameters
mu_delta <- 0.25
sigma_within <- 0.77
within_var <- sigma_within^2
alpha <- 0.05
groups <- k   # 5 groups

# search a wide enough range of n
n_seq <- n.small:400
tukey_power <- sapply(n_seq, function(n) {
  power.tukey.test(n = n, groups = groups, delta = mu_delta,
                   within.var = within_var, sig.level = alpha)$power
})

power_curve_tukey <- tibble(n = n_seq, power = tukey_power)

n_required_tukey <- n_seq[which(tukey_power >= 0.80)[1]]

ggplot(power_curve_tukey, aes(x = n, y = power)) +
  geom_line(linewidth = 0.6) +
  geom_hline(yintercept = 0.80, linetype = "dashed", color = "red") +
  geom_vline(xintercept = n_required_tukey, linetype = "dashed", color = "red") +
  labs(
    x = "Sample Size per Group (n)",
    y = "Power",
    title = "Power for Detecting a Pairwise Difference (Tukey HSD)",
    subtitle = paste0("k = ", groups, ", \u03B4 = ", mu_delta,
                      ", \u03C3 = ", sigma_within,
                      "; required n = ", n_required_tukey)
  ) +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.7),
    plot.title = element_text(face = "bold", size = 14)
  )
```



The required sample size for each group to achieve 80% on the pairwise comparison is n = 243. A sample size for each group of n = 40 achieves 80% for the overall ANOVA F-test, but not for a pairwise comparison due to the controlling of the FWER to 0.05 across all 10 comparisons. This increases the critical value and decreases alpha for each comparison, which in turn decreases power.

2.4 Part d

How does the answer to Part c change if they wanted to have adequate power for *all* pairwise comparisons? Use `Superpower::ANOVA_design()` to set up the ANOVA set up.

Note: Use design "5b" and the following.

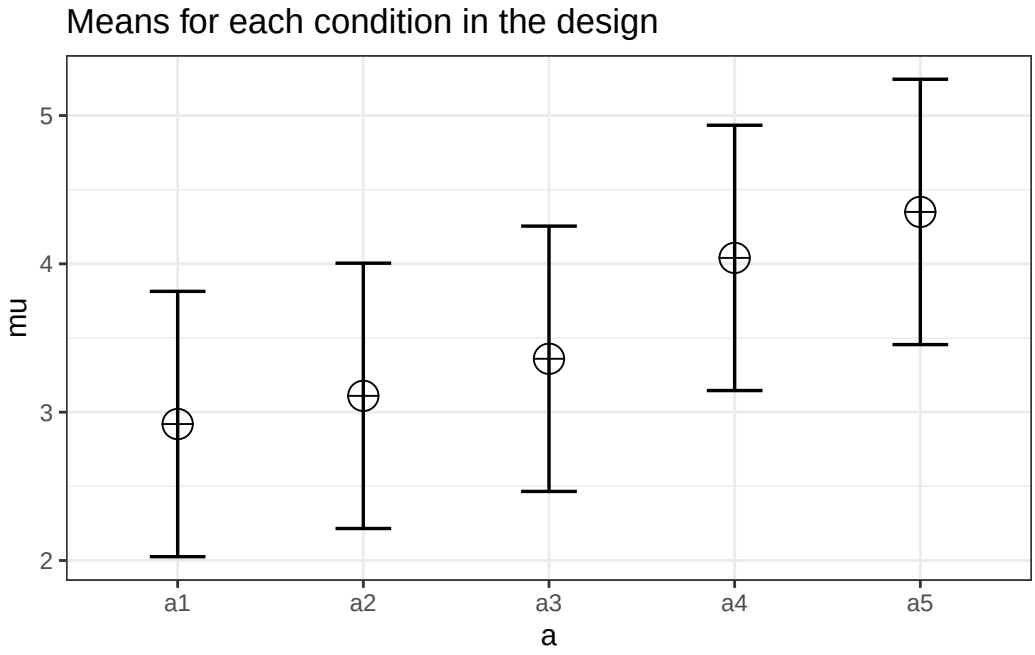
$$\mu = \{2.92, 3.11, 3.36, 4.04, 4.35\}$$
$$\sigma^2 = \frac{(97-1)0.87^2 + (176-1)0.85^2 + (135-1)0.84^2 + (67-1)1.00^2 + (67-1)0.90^2}{97+176+135+67+67-5} = 0.80$$

Compute the power of the researcher's design (part a) using `Superpower::ANOVA_power()`. If the researchers were able to oversample by double, a common approach to addressing this issue, would their power be at least 80% for all pariwise comparisons?

Note: This code can take a minute. You may want to set `cache: true` in your `R` chunk yaml header. You will also see more power because the actual effect size was 0.55.

```
# parameters
mu = c(2.92, 3.11, 3.36, 4.04, 4.35)
var = 0.80

# design setup using ANOVA_design()
design_result <- ANOVA_design(design = "5b",
                              n = 40,
                              mu = mu,
                              sd = sqrt(var))
```




```r
# power test
ANOVA_power(design_result = design_result,
            alpha_level = 0.05,
            nsims = 1000, # simulate 1000 times
            p_adjust = "bonferroni",
            seed = 1234)
```

```
Power and Effect sizes for ANOVA tests
        power effect_size
anova_a   100      0.2887

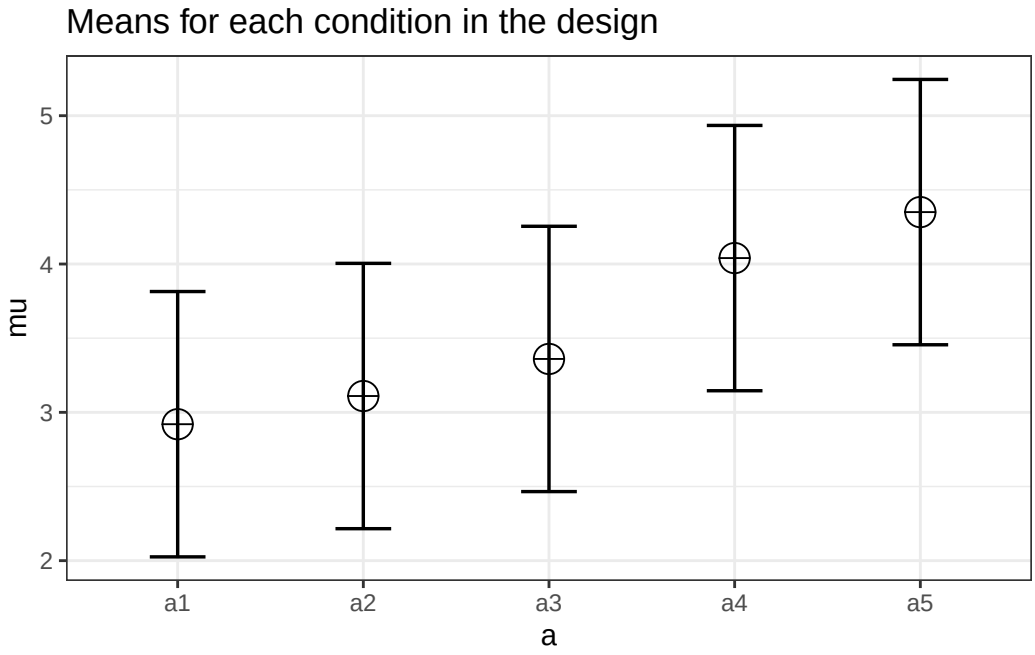
Power and Effect sizes for pairwise comparisons (t-tests)
            power effect_size
p_a_a1_a_a2   2.6      0.2125
p_a_a1_a_a3  25.2      0.4917
p_a_a1_a_a4  99.9      1.2608
p_a_a1_a_a5 100.0      1.6148
p_a_a2_a_a3   6.1      0.2810
p_a_a2_a_a4  95.6      1.0546
p_a_a2_a_a5 100.0      1.4096
p_a_a3_a_a4  70.1      0.7717
p_a_a3_a_a5  98.3      1.1243
p_a_a4_a_a5  11.2      0.3521

Power and Effect sizes for ANOVA tests
        power effect_size
anova_a   100      0.2887

Power and Effect sizes for pairwise comparisons (t-tests)
            power effect_size
p_a_a1_a_a2   2.6      0.2125
p_a_a1_a_a3  25.2      0.4917
p_a_a1_a_a4  99.9      1.2608
p_a_a1_a_a5 100.0      1.6148
p_a_a2_a_a3   6.1      0.2810
```

```
p_a_a2_a_a4  95.6      1.0546
p_a_a2_a_a5 100.0      1.4096
p_a_a3_a_a4  70.1      0.7717
p_a_a3_a_a5  98.3      1.1243
p_a_a4_a_a5  11.2      0.3521
```

```r
# doubling n
design_result_double <- ANOVA_design(design = "5b",
                                     n = 80,
                                     mu = mu,
                                     sd = sqrt(var))
```




```r
# power test for double n
ANOVA_power(design_result = design_result_double,
            alpha_level = 0.05,
            nsims = 1000,
            p_adjust = "bonferroni",
            seed = 1234)
```

```
Power and Effect sizes for ANOVA tests
        power effect_size
anova_a   100      0.2803

Power and Effect sizes for pairwise comparisons (t-tests)
            power effect_size
p_a_a1_a_a2   7.2      0.2118
p_a_a1_a_a3  61.5      0.4936
p_a_a1_a_a4 100.0      1.2551
p_a_a1_a_a5 100.0      1.6036
p_a_a2_a_a3  13.4      0.2817
p_a_a2_a_a4  99.9      1.0445
p_a_a2_a_a5 100.0      1.3942
p_a_a3_a_a4  98.1      0.7641
p_a_a3_a_a5 100.0      1.1134
p_a_a4_a_a5  26.6      0.3484
```

```
Power and Effect sizes for ANOVA tests
        power effect_size
anova_a   100      0.2803

Power and Effect sizes for pairwise comparisons (t-tests)
            power effect_size
p_a_a1_a_a2   7.2      0.2118
p_a_a1_a_a3  61.5      0.4936
p_a_a1_a_a4 100.0      1.2551
p_a_a1_a_a5 100.0      1.6036
p_a_a2_a_a3  13.4      0.2817
p_a_a2_a_a4  99.9      1.0445
p_a_a2_a_a5 100.0      1.3942
p_a_a3_a_a4  98.1      0.7641
p_a_a3_a_a5 100.0      1.1134
p_a_a4_a_a5  26.6      0.3484
```

Even if the researchers doubled their sample size per group to n = 80, they will not achieve 80% across all pairwise comparisons.

3 Question 3: An Application

Yang et al. (2023) investigate whether low-dose nicotine can counteract age-related metabolic decline. The proposed mechanism is through NAD^+, which is a chemical compound essential for metabolic energy production. The idea is that low-dose nicotine could help improve and/or maintain the recycling of NAD^+ by activating the salvage enzyme NAMPT via SIRT1. This could, in turn, restore NAD^+ homeostasis, which is the biological balance maintained within a cell between the production, recycling, and degradation of nicotinamide adenine dinucleotide (NAD^+).

Note: It is low-dose nicotine. Cigarette smoking delivers massive doses of nicotine along with other harmful chemicals. This is not an endorsement for smoking.

Male mice across distinct life stages – 2, 6, and 16 months – were categorized to evaluate natural, age-dependent metabolic and enzymatic shifts in two conditions.

- Treatment: low-dose nicotine (administered in drinking water)
- Control: no-dose placebo (administered in drinking water)

The researchers then aimed to compare treatment against control in the following samples:

- 2 months old, control
- 6 months old, control
- 16 months old, control
- 16 months old, treatment

Using PET imaging, the researchers measured age-induced inflammatory glucose hypermetabolism (SUV_{max}). They want to show that this quantity increases with age and those in the nicotine group saw levels comparable to younger mice.

Table 1: SUV_{max} measurements in the heart across age and treatment groups

Replicate	2M	6M	16M	16M, Treatment
1	1.48969	4.36000	7.30182	4.69304
2	1.30168	4.28807	7.12416	2.51326
3	1.30620	3.64184	8.28764	2.38019

3.1 Part a

State the null (H_0) and alternative (H_a) hypotheses for the omnibus test.

H_0: means are = for all groups H_a: at least one mean is different

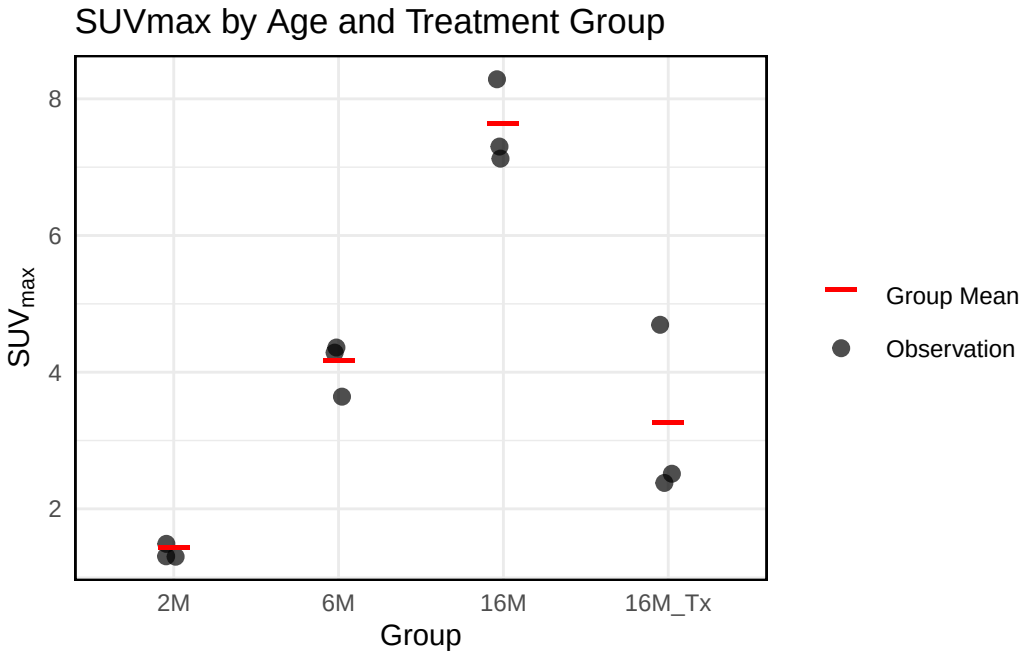
3.2 Part b

Perform some exploratory data analysis to provide empirical information about the researchers question.

```
# reshape SUVmax data into long format as one row per observation
suv_long <- tibble(
  group = factor(rep(c("2M", "6M", "16M", "16M_Tx"), each = 3),
                 levels = c("2M", "6M", "16M", "16M_Tx")),
  suvmax = c(1.48969, 1.30168, 1.30620,
             4.36000, 4.28807, 3.64184,
             7.30182, 7.12416, 8.28764,
             4.69304, 2.51326, 2.38019)
)

# compute the mean SUVmax within each group
group_means <- suv_long |>
  group_by(group) |>
  summarize(mean_suv = mean(suvmax), .groups = "drop")

ggplot() +
  geom_jitter(data = suv_long, aes(x = group, y = suvmax, color = "Observation"),
              width = 0.05, size = 2.5, alpha = 0.7) +
  geom_point(data = group_means,
             aes(x = group, y = mean_suv, color = "Group Mean"),
             shape = "-", size = 8) +
  scale_color_manual(values = c("Observation" = "black", "Group Mean" = "red")) +
  labs(
    x = "Group",
    y = expression(SUV[max]),
    title = "SUVmax by Age and Treatment Group",
    color = NULL
  ) +
  theme_minimal() +
  theme(panel.border = element_rect(color = "black", fill = NA, linewidth = 0.7))
```



Just from the exploratory data, I can see the 16M treatment group mean looks close to the 6M group mean, and potentially the 2M group mean, but very different from the 16M control group mean.

3.3 Part c

Perform and report the results of the omnibus test. Make sure to evaluate the conditions for using the method you choose.

```
(test.result <- oneway.test(suvmax ~ group, data = suv_long, var.equal = FALSE))
```

```
    One-way analysis of means (not assuming equal variances)

data:  suvmax and group
F = 98.464, num df = 3.0000, denom df = 3.5496, p-value = 0.0006851
```

Check conditions fo ANOVA: 1. The SUVmax is quantitative. 2. The population distributions are approximately Gaussian. 3. The population distributions do NOT have the same variance. 4. The sample size is n = 3 across all groups. 5. The sample observations are independent.

The variance condition isn't met, so check conditions for Welch's ANOVA. 1. The SUVmax is quantitative. 2. The population distributions are approximately Gaussian. 3. The sample sizes are n = 3 across all groups. 4. The sample is representative across the population 5. The sample observations are independent. 6. The samples are independent.

After using a one-way Welch's ANOVA test, not assuming equal variances, we get a p-value of around 0.00069 $< \alpha = 0.05$. We have statistically discernible support that the mean SUVmax varies among at least one group.

3.4 Part d

Conduct post-hoc pairwise comparisons to determine if age-induced inflammatory glucose hypermetabolism increase with age and whether low-dose nicotine reduces age-induced inflammatory glucose hypermetabolism.

```r
posthoc.result <- games_howell_test(suvmax ~ group, data = suv_long) |>
  mutate(Comparison = paste(group2, "-", group1)) |>
  select(Comparison, estimate, conf.low, conf.high, p.adj)

posthoc.result
```

```
# A tibble: 6 x 5
  Comparison   estimate conf.low conf.high   p.adj
  <chr>           <dbl>    <dbl>     <dbl>   <dbl>
1 6M - 2M          2.73     1.31      4.15  0.0119
2 16M - 2M         6.21     3.81      8.60  0.00720
3 16M_Tx - 2M      1.83    -3.30      6.96  0.308
4 16M - 6M         3.47     1.56      5.39  0.00801
5 16M_Tx - 6M    -0.901    -5.47      3.67  0.698
6 16M_Tx - 16M   -4.38     -8.51     -0.244 0.0431
```

3.5 Part e

Write a short paragraph directly addressing the researcher's hypothesis using your work in (a)-(d).

After conducting a Welch's ANOVA test, we found statistically discernible support that at least one group (2M, 6M, 16M, and 16M_Tx) had a different mean SUVmax from the others ($p = 0.00069$). After conducting a post-hoc analysis using a Games Howell test for multiple comparisons, we found that, among the 3 control groups, the mean SUVmax in male mice increases from 2 months to 6 months to 16 months ($p = 0.012$ and $p = 0.008$, respectively). Additionally, we found that the 16 month treatment group experienced a discernibly lower SUVmax than the 16 month control group ($p = 0.043$) and that it was comparable to the SUVmax of the 2 month group ($p = 0.308$) and the 6 month group ($p = 0.698$). Overall, our data support that SUVmax increases with age, and that a low-dose nicotine treatment can lower SUVmax in older mice to levels similar to that of younger mice. A major limitation to consider from this study is that we only observed 3 samples per group; further research should be aimed at acquiring more samples to increase the statistical power of the study.

References

Yang, Liang, Junfeng Shen, Chunhua Liu, et al. 2023. "Nicotine Rebalances NAD+ Homeostasis and Improves Aging-Related Symptoms in Male Mice by Enhancing NAMPT Activity." *Nature Communications* 14 (1): 900.